

Supplementary Material

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I. ADDITIONAL METHOD DETAILS

A. Keypoint Annotation Details

For each object category, we manually annotate $N_p = 16$ semantic keypoints in the canonical object coordinate space. These keypoints are selected to capture pose-discriminative local structures. For example, for the bottle category we annotate the bottle mouth, the bottom, and the neck region; for the mug category we annotate the handle attachment points, the rim, and the bottom. Notably, the annotation process is lightweight and does not require careful placement of each keypoint. The locations are determined based on intuitive geometric understanding of the category, taking approximately one minute per category. The robustness to imprecise annotation is enabled by the Prior Adapter, which projects the extracted ULIP-2 [1] features into the network feature space through learnable linear projections. During training, the Prior Adapter adapts the prior features via gradient backpropagation, effectively compensating for annotation inaccuracies and enabling the network to discover pose-relevant patterns beyond the manually specified locations. Fig. 1 shows examples of the annotated keypoints across all categories used in our experiments.

B. Local-Cross-Global Attention Block

The Local-Cross-Global (LCG) Attention Block refines the group-wise keypoint features $\{\mathbf{F}_{kp}^g\}_{g=1}^G$ through three sequential stages. We provide the detailed formulations here.

Local Refinement: For each group g , let $\mathbf{P}_{kp}^g \in \mathbb{R}^{N_g \times 3}$ and $\mathbf{F}_{kp}^g \in \mathbb{R}^{N_g \times C}$ denote the keypoint coordinates and features. We retrieve the $K = 16$ nearest object points for each keypoint, yielding neighbor coordinates $\mathbf{P}_{\mathcal{N}}^g \in \mathbb{R}^{N_g \times K \times 3}$ and neighbor features $\mathbf{F}_{\mathcal{N}}^g \in \mathbb{R}^{N_g \times K \times C}$. The relative positional encoding is computed as:

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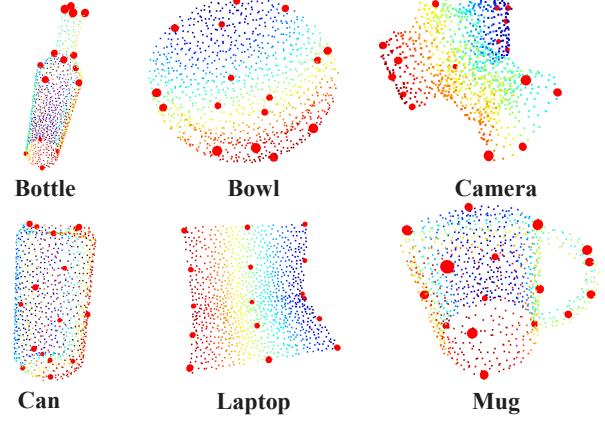


Fig. 1. Examples of manually annotated category-level keypoints used to extract keypoint feature priors. Each red colored dot represents a semantic keypoint location in the canonical object space.

$$\Delta \mathbf{P}_{loc}^g = \phi_{loc}(\mathbf{P}_{\mathcal{N}}^g - \mathbf{P}_{kp}^g), \quad (1)$$

where $\mathbf{P}_{kp}^g$ is broadcast along the neighbor dimension, and $\phi_{loc}(\cdot)$ is a shared MLP that maps 3D offsets to C -dimensional positional encodings. The locally refined feature is obtained by cross-attention:

$$\mathbf{F}_{loc}^g = \text{CrossAttn}(\mathbf{F}_{kp}^g, \mathbf{F}_{\mathcal{N}}^g + \Delta \mathbf{P}_{loc}^g), \quad (2)$$

where the query is $\mathbf{F}_{kp}^g$ and the key-value pair is the position-enhanced neighbor features.

Cross-Group Interaction: After local refinement, we concatenate all group-wise features as $\mathbf{F}_{loc} \in \mathbb{R}^{GN_g \times C}$. For each group g , the query is its own features $\mathbf{F}_{loc}^g \in \mathbb{R}^{N_g \times C}$, and the key-value pair is the features from all *other* groups:

$$\mathbf{F}_{cross}^g = \text{CrossAttn}(\mathbf{F}_{loc}^g, \mathbf{F}_{loc}^{\setminus g}), \quad (3)$$

where $\mathbf{F}_{loc}^{\setminus g} \in \mathbb{R}^{(G-1)N_g \times C}$ denotes features from groups $\{1, \dots, G\} \setminus \{g\}$. This design enables each group to leverage complementary cues discovered by other groups while maintaining group-specific specialization.

Global Aggregation: Finally, we encode intra-group structural relations. A global positional embedding is computed for each group based on pairwise relative positions:

$$\tilde{\mathbf{P}}_{glob}^g = \mathbf{E}_{pos}^g + \frac{1}{N_g} \sum_{j=1}^{N_g} \phi_{glob}(\mathbf{P}_{kp}^g - \mathbf{p}_{kp,j}^g), \quad (4)$$

where $\mathbf{E}_{pos}^g \in \mathbb{R}^{N_g \times C}$ is a learnable absolute positional embedding matrix, $\mathbf{p}_{kp,j}^g \in \mathbb{R}^3$ is the j -th keypoint coordinate (broadcast for subtraction), and $\phi_{glob}(\cdot)$ is an MLP that encodes pairwise offsets into C -dimensional embeddings. A group-level context vector is obtained by average pooling:

$$\bar{\mathbf{F}}_{cross}^g = \frac{1}{N_g} \sum_{i=1}^{N_g} \mathbf{f}_{cross,i}^g, \quad (5)$$

which is broadcast to all N_g keypoints when used. The updated keypoint feature is computed via a residual MLP:

$$\hat{\mathbf{F}}^g = \text{MLP} \left([\mathbf{F}_{cross}^g, \bar{\mathbf{F}}_{cross}^g, \tilde{\mathbf{P}}_{glob}^g] \right) + \mathbf{F}_{cross}^g, \quad (6)$$

where $[\cdot, \cdot, \cdot]$ denotes channel-wise concatenation. The final output of the LCG Block is obtained by applying self-attention with layer normalization:

$$\mathbf{F}_{out}^g = \hat{\mathbf{F}}^g + \text{SelfAttn} \left(\text{LN} \left(\hat{\mathbf{F}}^g \right) \right), \quad (7)$$

C. Corner Point Supervision

To encourage the learned keypoints to cover geometrically meaningful regions beyond simple surface sampling, we introduce corner point supervision that utilizes image-level mask information. As shown in Fig. 2, given the segmentation mask of the target object in the RGB image, we extract the 2D contour of the mask. We then sample $N_c = 10$ corner points along this contour. These 2D corner points are then back-projected to 3D space using the depth map and camera intrinsics, yielding a set of reconstructed 3D corner points $\mathbf{Q}_c = \{\mathbf{q}_j\}_{j=1}^{N_c}$.

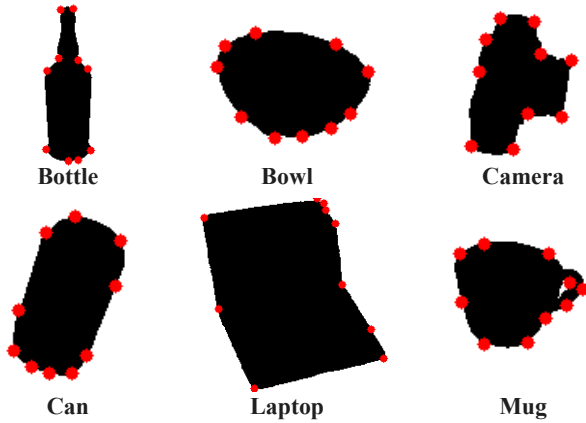


Fig. 2. Visualization of corner points extracted from the mask image.

D. Diversity Loss Formulation

The diversity loss $\mathcal{L}_{div}$ prevents keypoint collapse at two levels.

a) *Within-group diversity*.: For each group g , we encourage a minimum separation τ between any pair of keypoints within the same group. Let $d_{ij}^g = \|\mathbf{p}_{kp,i}^g - \mathbf{p}_{kp,j}^g\|_2$ denote the Euclidean distance between the i -th and j -th keypoints in group g . The within-group loss is:

$$\mathcal{L}_{within} = \sum_{g=1}^G \frac{1}{GN_g(N_g - 1)} \sum_{i=1}^{N_g} \sum_{\substack{j=1 \\ j \neq i}}^{N_g} \left[1 - \frac{d_{ij}^g}{\tau} \right]_+. \quad (8)$$

where $[x]_+ = \max(0, x)$.

This formulation penalizes keypoint pairs whose distance falls below τ , with a normalized penalty that equals 1 when $d_{ij}^g = 0$ and decays linearly to 0 when $d_{ij}^g \geq \tau$.

b) *Between-group diversity*.: For keypoints with index i across different groups, we apply a two-part penalty to encourage both a minimum separation τ and an extra incentive to exceed a larger threshold τ_1 (where $\tau_1 = 0.5\tau$). Let $d_i^{g,h} = \|\mathbf{p}_{kp,i}^g - \mathbf{p}_{kp,i}^h\|_2$ denote the distance between the i -th keypoints in groups g and h . The between-group loss is:

$$\mathcal{L}_{between} = \frac{1}{G(G-1)} \sum_{g=1}^G \frac{1}{N_g} \sum_{i=1}^{N_g} \sum_{\substack{h=1 \\ h \neq g}}^G \left(\left[1 - \frac{d_i^{g,h}}{\tau} \right]_+ + \sigma \left(\frac{d_i^{g,h}}{\tau_1} - 1 \right) - \frac{1}{2} \right). \quad (9)$$

where $\sigma(\cdot)$ denotes the sigmoid function and $[x]_+ = \max(0, x)$. The first term penalizes distances below τ , following the same normalized form as $\mathcal{L}_{within}$. The second term provides a smooth, bounded penalty when $d_i^{g,h}$ exceeds τ_1 : the sigmoid activation saturates at approximately 0.5 for large distances, preventing unbounded gradient growth while still encouraging separation beyond τ_1 . The overall diversity loss is the average of both terms:

$$\mathcal{L}_{div} = \frac{1}{2} (\mathcal{L}_{within} + \mathcal{L}_{between}). \quad (10)$$

II. ADDITIONAL ABLATION DETAILS

A. Keypoint Gradient Uniformity Analysis

To better explain the gradient uniformity results, we describe how the four metrics are computed. A simple L1 loss is used to supervise the predicted pose [2]–[4], including rotation, translation, and size:

$$\mathcal{L}_{pose} = \|\mathbf{R} - \mathbf{R}_{gt}\|_1 + \|\mathbf{t} - \mathbf{t}_{gt}\|_1 + \|\mathbf{s} - \mathbf{s}_{gt}\|_1, \quad (11)$$

where $(\mathbf{R}, \mathbf{t}, \mathbf{s})$ and $(\mathbf{R}_{gt}, \mathbf{t}_{gt}, \mathbf{s}_{gt})$ denote the predicted and ground-truth rotation, translation, and size, respectively.

All metrics are computed using all valid object instances in the REAL275 test set. Since the semantic keypoints are defined in a category-specific canonical space, directly averaging the same keypoint index across different categories may mix different semantic meanings. Therefore, we first compute the gradient statistics within each object category and then average the resulting metrics across categories.

During backpropagation, we compute the gradient of $\mathcal{L}_{\text{pose}}$ with respect to each predicted 3D keypoint. For the m -th object instance in category c and the k -th keypoint, the gradient norm is defined as

$$g_{m,k}^c = \left\| \frac{\partial \mathcal{L}_{\text{pose}}}{\partial \mathbf{p}_{m,k}^c} \right\|_2, \quad (12)$$

where $\mathbf{p}_{m,k}^c$ is the predicted 3D coordinate of the k -th keypoint for category c . We then average the gradient norm over all valid instances within the same category:

$$\bar{g}_k^c = \frac{1}{M_c} \sum_{m=1}^{M_c} g_{m,k}^c, \quad (13)$$

where M_c is the number of valid object instances of category c . The category-wise gradient vector is written as

$$\bar{\mathbf{g}}^c = [\bar{g}_1^c, \bar{g}_2^c, \dots, \bar{g}_K^c], \quad (14)$$

where K is the number of keypoints for each category.

Based on $\bar{\mathbf{g}}^c$, we compute the four gradient uniformity metrics for each category independently. First, the Gini coefficient measures the imbalance of the gradient distribution. Let $\bar{g}_{(k)}^c$ be the k -th value after sorting $\bar{\mathbf{g}}^c$ in ascending order. The category-wise Gini coefficient [5], [6] is computed as

$$\text{Gini}^c = \frac{2 \sum_{k=1}^K k \bar{g}_{(k)}^c}{K \sum_{k=1}^K \bar{g}_{(k)}^c} - \frac{K+1}{K}. \quad (15)$$

A lower Gini coefficient indicates a more balanced gradient distribution among keypoints.

Second, the coefficient of variation is computed as

$$\text{CV}^c = \frac{\text{Std}(\bar{\mathbf{g}}^c)}{\text{Mean}(\bar{\mathbf{g}}^c) + \epsilon}, \quad (16)$$

where ϵ is a small constant for numerical stability. A lower CV [7] indicates smaller relative variation among keypoint gradients.

Third, the maximum-to-minimum gradient ratio is defined as

$$\rho^c = \frac{\max_k \bar{g}_k^c}{\min_k \bar{g}_k^c + \epsilon}. \quad (17)$$

A smaller ratio indicates that the gradient distribution is less dominated by a few keypoints.

Finally, the effective keypoint count is computed by first normalizing the category-wise gradient norms [8]:

$$p_k^c = \frac{\bar{g}_k^c}{\sum_{j=1}^K \bar{g}_j^c + \epsilon}, \quad (18)$$

and then calculating

$$K_{\text{eff}}^c = \exp \left(- \sum_{k=1}^K p_k^c \log(p_k^c + \epsilon) \right). \quad (19)$$

This value estimates how many keypoints effectively receive pose-related gradients. If the gradients are uniformly distributed over all K keypoints, then K_{eff}^c approaches K ; if the gradients are concentrated on only a few keypoints, then K_{eff}^c becomes smaller.

The final reported value of each metric is obtained by averaging the category-wise metric over all evaluated categories:

$$\text{Metric} = \frac{1}{N_{\text{cat}}} \sum_{c=1}^{N_{\text{cat}}} \text{Metric}^c, \quad (20)$$

where N_{cat} is the number of object categories. Lower Gini, CV, and maximum-to-minimum ratio indicate a more uniform gradient distribution, while a larger K_{eff} means that more keypoints receive effective pose-related gradients.

REFERENCES

- [1] L. Xue, N. Yu, S. Zhang, J. Li, R. Mart'in-Mart'in, J. Wu, C. Xiong, R. Xu, J. C. Niebles, and S. Savarese, "Ulip-2: Towards scalable multimodal pre-training for 3d understanding," *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 27 081–27 091, 2023.
- [2] X. Lin, Y. Peng, L. Wang, X. Zhong, M. Zhu, J. Yang, C. Liu, and Q. Chen, "Cleanpose: Category-level object pose estimation via causal learning and knowledge distillation," *ArXiv*, vol. abs/2502.01312, 2025.
- [3] Y. Yang, P. Song, E. Lan, D. Liu, and J. Liu, "Mk-pose: Category-level object pose estimation via multimodal-based keypoint learning," *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4232–4239, 2025.
- [4] X. Lin, W. Yang, Y. Gao, and T. Zhang, "Instance-adaptive and geometric-aware keypoint learning for category-level 6d object pose estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 21 040–21 049, 2024.
- [5] C. Gini, *Variabilità e mutabilità: contributo allo studio delle distribuzioni e delle relazioni statistiche.* [Fasc. I.]. Tipogr. di P. Cuppini, 1912.
- [6] F. Cowell, *Measuring inequality*. Oxford University Press, 2011.
- [7] G. F. Reed, F. Lynn, and B. D. Meade, "Use of coefficient of variation in assessing variability of quantitative assays," *Clinical and Vaccine Immunology*, vol. 9, no. 6, pp. 1235–1239, 2002.
- [8] L. Jost, "Entropy and diversity," *Oikos*, vol. 113, no. 2, pp. 363–375, 2006.



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